Concept unlearning in variational autoencoders

Measuring residual information in latent representations

The University of Queensland

Project Proposal

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1. Introduction

Under the General Data Protection Regulation (GDPR), Article 17 establishes the "right to erasure," requiring organisations to delete personal data upon request under certain conditions. Additionally, Article 22 places limits on fully automated decision-making based on such data [5]; however, after raw data is deleted, trained models still retain implicit information about individuals. This raises questions about whether and how models must be modified, motivating research into machine unlearning.

Unlearning ensures that a trained model no longer reflects the contribution of a designated subset of data, without requiring full retraining from scratch [2]. Key literature distinguishes parametric unlearning, modifying model parameters to approximate a retrained oracle [3], from surface-level unlearning, matching the oracle's output without altering internal representations [4, 10]. Whether output-level methods truly eliminate the underlying representations of a removed concept remains an open question. Variational Autoencoders (VAEs) provide a natural setting in which to study this: their encoders map inputs to an explicit latent distribution, making the structure of learned representations directly measurable in a way that is not straightforwardly available in discriminative or purely generative models. My central guiding question is:

> *To what extent does the change in the geometry of a VAE's latent space predict the amount of residual concept information retained in the model, independent of its output behaviour?*

1.1 Project Scope and Topic Definition

A broader open question in machine unlearning is whether methods that suppress a concept at the output level also remove its internal representation within the model. Answering this requires an architecture with an explicit, measurable latent space; one where representation geometry can be directly analysed rather than inferred from outputs alone. This project uses VAEs as a concrete and well-understood setting for investigating that question. VAEs impose a structured probabilistic latent space that enables the application of geometric metrics — such as probing classifiers, CKA, and cluster analysis — directly to the encoder's representations, making them particularly suitable for this kind of evaluation.

Concretely, this project examines concept unlearning in VAEs, investigating whether unlearning methods that suppress a visual concept at the output level also remove its internal representation in the model's latent space. The forget concept is eyeglasses on the CelebA face-attribute dataset [14] as the primary setting, and heart shapes on the dSprites synthetic dataset [1] as a controlled validation setting. The project is bounded to the VAE architecture and does not extend to GANs, diffusion models, or discriminative classifiers. Unlearning is evaluated at the level of the encoder's latent representations, not solely at the decoder output, using a multi-metric framework comprising probing classifiers, CKA, cluster analysis, k-NN graph connectivity, and adversarial recovery.

1.2 Research Questions

1. Do different unlearning methods produce different changes to the VAE's latent manifold geometry for the forgotten concept?
2. Do those geometric changes predict the degree of residual concept information; i.e., how well the concept can be recovered via probing or adversarial perturbation?
3. How closely does each method's effect on latent geometry approximate that of the oracle, and does intervention depth predict the size of that gap?

1.3 Contributions

The primary contributions of this project are:

1. **A representation-level evaluation framework for generative unlearning.** Existing evaluations of concept unlearning predominantly measure success at the output level, leaving open whether internal representations are affected. This project provides a systematic measurement of whether unlearning methods alter the latent geometry of a VAE, using CKA, probing, cluster analysis, and k-NN connectivity as complementary metrics.
2. **A comparative taxonomy of unlearning methods by intervention depth.** This project organises naive, representation-level, and weight-level methods into a principled taxonomy and empirically tests whether intervention depth predicts the degree of internal forgetting — a question left open by existing surveys [3].
3. **A dual-dataset validation design.** By conducting all experiments on both CelebA (entangled, real-world attributes) and dSprites (fully independent generative factors), this project isolates the effect of attribute entanglement on unlearning completeness — a confound that prior work identifies but does not control for [16].

2. Background

2.1 Machine Unlearning

Bourtoule et al. define the machine unlearning problem as follows: given a model trained on dataset D and a forget set $D_f \subset D$, is it possible to produce a model indistinguishable from one retrained from scratch on $D \setminus D_f$ [2]? This *exact* unlearning criterion is the gold standard but is computationally prohibitive for large neural networks, motivating *approximate* methods reducing the influence of D_f through parameter updates or fine-tuning. This sacrifices formal guarantees in favour of efficiency [2, 19].

2.2 The Verification Problem

Golatkar et al. establish the central verification problem motivating this project [8]. They demonstrate naive unlearning methods — fine-tuning on the retain set, applying random labels to the forget set, removing the relevant output neuron — all fail under weight-probing attacks: a probing function applied to the model's weights recovers forgotten information even when output behaviour appears correct. To address this, they propose scrubbing via the Fisher Information Matrix (FIM), injecting structured noise in FIM-weighted directions to push the weight distribution toward the oracle's [8]. The re-learn time after scrubbing matches the oracle, whereas naive methods recover performance in very few epochs.

Hsu et al. extend this finding from weights to latent neighbourhoods [10]. Even when an unlearned model's accuracy on forget samples matches the oracle, residual knowledge persists nearby: slightly perturbed forget samples are recognised by the unlearned model but not by the oracle. In a VAE, where the latent space is a continuous manifold and nearby points encode semantically similar content, this neighbourhood-level residual is a direct privacy vulnerability; an adversary can traverse the latent space from a retain-set point toward the forgotten region without triggering output-level detection.

2.3 Variational Autoencoders and Latent Geometry

A VAE consists of an encoder $q_\phi(z|x)$, which maps input x to a distribution over latent codes z, and a decoder $p_\theta(x|z)$, which reconstructs x from sampled z. Training maximises the Evidence Lower Bound (ELBO): $\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x) \| p(z))$, balancing reconstruction fidelity against regularisation toward a standard Gaussian prior [12].

Mathieu et al. show that the VAE latent space carries a natural Riemannian metric induced by the decoder, where visual concepts correspond to compact submanifolds [15]. This is the theoretical basis for this project's evaluation framework: genuine concept erasure should dissolve the forget-concept submanifold, reducing its compactness and separability. Surface-level suppression leaves it geometrically intact despite altered decoder outputs.

Higgins et al. introduce β-VAE, weighting the KL term by $\beta > 1$ to encourage axis-aligned disentanglement [9]. In a more disentangled model, a target attribute occupies fewer latent dimensions, making removal more surgical and reducing collateral damage to retained attributes. This project includes a $\beta = 4$ variant to test whether disentanglement facilitates more thorough

unlearning, using dSprites — the benchmark introduced alongside β-VAE, with fully independent generative factors — as the idealised baseline for this comparison.

2.4 Feature Unlearning in Generative Models

Moon et al. provide the most directly relevant prior work, proposing a feature unlearning method for pre-trained VAEs and GANs evaluated on CelebA attributes including eyeglasses [16]. Their approach identifies the latent direction encoding the target attribute via a linear probe, then fine-tunes the model to suppress that direction while preserving reconstruction quality on non-feature images. The method demonstrates output-level robustness to adversarial attacks, but does not measure whether the latent geometry is altered. This is the gap this project fills: by applying Moon et al.'s method alongside naive and weight-level approaches, and evaluating all of them against a common set of latent-space metrics, this project determines whether output-level success implies internal forgetting.

Across related work in StyleGAN [18] and diffusion models [7, 17], the same pattern holds: behavioural suppression does not imply internal forgetting. This convergence across architectures confirms that a representation-level evaluation framework is necessary, and that the VAE setting studied here is an important and underexamined case.

3. Literature Review

This literature review is organised into four sections:

- **Machine Unlearning: Exact vs. Approximate** — definitions, motivations, and key methods including SISA, parametric updates, and influence functions.
- **Generative Model Unlearning** — unlearning in VAEs and GANs, with a focus on feature removal and latent manipulation.
- **Representation Geometry and Evaluation Tools** — latent space analysis using CKA, probing, adversarial recovery, and clustering methods.
- **Synthesis and Open Challenges** — key insights, limitations, and open problems in evaluating unlearning.

3.1 Machine Unlearning: Exact vs. Approximate

3.1.1 Motivations and Formal Definitions

Exact unlearning requires the unlearned model to be statistically indistinguishable from a model trained from scratch without the target data. This gold standard is computationally prohibitive for large deep networks, birthing a rich landscape of approximate methods. The approximation criterion is typically grounded in Differential Privacy (DP): an unlearned model is acceptable if it remains within a DP-style distance from the oracle retrained model [2]. Contemporary surveys note that unlearning methods are split into retraining-based approaches and post-hoc parameter updates, with further subdivision by whether they provide formal guarantees or rely on empirical evaluation alone [3].

3.1.2 SISA: Data Partitioning for Exact Unlearning

Bourtoule et al. introduced Sharded, Isolated, Sliced, and Aggregated (SISA) training — the most cited framework for achieving near-exact unlearning without fully retraining from scratch. The core insight is architectural: SISA partitions the training set into disjoint shards and trains a sub-model on each, so that when an unlearning request arrives, only the affected shard requires retraining [2]. However, SISA has been evaluated almost entirely on discriminative models and is out of scope for this project: its shard-based design assumes aggregatable outputs, whereas independently-trained VAE generators lack a shared latent space and require architectural changes to build a coherent model [2].

3.1.3 Fisher Information and Gradient-Based Unlearning

Golatkar et al. introduced the scrubbing framework in"Eternal Sunshine of the Spotless Net: Selective Forgetting in Deep Networks." Their central insight is directly relevant to this project: at the output level, the effects of forgotten data is concealable; however, it remains recoverable from the weights. They demonstrated how naive methods (fine-tuning on the retain set, applying

random labels to the forget set, removing the relevant output neuron) all fail under weight-probing attacks: a probing function applied to the weights can recover the forgotten information even when output behaviour appears correct [8].

To remedy this, they proposed a principled scrubbing procedure derived from the Fisher Information Matrix (FIM). The FIM characterises which weight directions encode information about a particular training cohort. By adding structured Gaussian noise in FIM-weighted directions — large enough to mask the forget set's imprint but small enough not to disrupt retained knowledge — the scrubbed model's weight distribution is pushed closer in KL divergence to that of the oracle [8]. Empirical results showed the re-learn time (epochs needed to recover the forgotten class) after scrubbing matches the oracle, whereas naive methods recover performance in very few epochs. This method is included as a stretch goal in this project.

The negative gradient approach — maximising the loss on the forget set via gradient ascent — is a simpler parametric method that I include as a method. Golatkar et al. found that, like other naive methods, it leaves residual information recoverable through probing [8]. More recent work such as "Representation Misdirection for Unlearning" [11] showed that steering internal activations of forget samples toward random representations is more effective than gradient ascent alone.

3.1.4 Influence Functions and Data Valuation

Influence function approaches estimate the effect of removing a single training example by approximating the change in model parameters via a second-order Taylor expansion of the training loss. Koh et al. demonstrated that influence functions can identify the most influential training for a model's predictions; and, removing high-influence examples changes outputs in predictable directions. In an unlearning context, influence functions are a computationally efficient proxy for the parameter change that would result from retraining without target examples. The Influence-guided Machine Unlearning method (IMU) extends this to efficiently estimate and apply the necessary parameter updates [6].

3.1.5 Surface-Level vs. Parametric Unlearning

This project investigates the distinction between surface-level behavioural suppression and genuine parametric forgetting. Surface-level methods modify only what the model outputs for the target concept, leaving the internal encoding unchanged. These approaches are computationally cheap and easy to deploy, but they are fragile; adversarial prompts, fine-tuning with the original data, or by probing the internal representations directly can circumvent it [3].

Hsu et al. formalises this fragility in "Are We Really Unlearning? The Presence of Residual Knowledge." Even when an unlearned model's accuracy on forget samples matches the oracle's, residual knowledge persists in the vicinity of forgetting samples. Specifically, slightly perturbed versions of a forget sample are recognised by the unlearned model but not by the oracle. This residual knowledge is a privacy vulnerability: any adversary can generate or obtain nearby samples to extract information about the forgotten data [10]. The difference between direct knowledge removal and neighbourhood-level removal is important for VAEs, where the latent space is a continuous manifold and nearby points encode semantically similar content.

My project investigates this contrast through two unlearning tasks: eyeglasses on CelebA as the primary setting, and heart shapes on dSprites as a controlled validation setting where generative factors are known and the forget concept is cleanly separable by design. If a method truly

erases a concept, forget-set images will reconstruct incorrectly and latent-space perturbations starting from retain-set images cannot access the forgotten region.

3.2 Generative Model Unlearning

3.2.1 The Moon et al. Feature Unlearning

Moon et al. provide the most directly relevant prior work for this project in "Feature Unlearning for Pre-trained GANs and VAEs." Their framework removes a specific visual attribute from a pre-trained VAE and evaluates performance on CelebA across attributes such as 'bangs', 'beards', and 'eyeglasses' [16].

Moon et al. frame feature unlearning as distinct from the standard data removal problem. Rather than removing specific images from the training set, the goal is to remove a visual concept that may appear across many training images, each also containing other attributes that should be preserved; unlearning images containing glasses risks degrading generation of all other facial features present in those images.

A key finding motivating this project is their adversarial robustness evaluation: models unlearned using this method were more resistant to adversarial attacks than naive fine-tuning, but Moon et al. did not quantify whether the latent geometry was altered, only that the outputs improved [16]. This is precisely the gap my project fills: by applying Moon et al.'s method alongside naive and weight-level approaches and evaluating all of them against a common set of latent-space metrics, this project determines whether output-level success implies internal forgetting.

3.2.2 GAN-Related Work and Methodological Context

Several recent works on generative model unlearning provide useful context for this project, despite not addressing VAEs directly. In the StyleGAN literature, Sun et al. proposed a cascaded unlearning method in latent space, where glasses-related directions in the W^+ space are identified and neutralised by substituting nearby non-glasses latent codes. Their method is effective at the output level, as generated images no longer contain glasses, but the underlying latent structure remains intact, with glasses-related directions still present but encoded differently through the modified generator [18].

Diffusion-model unlearning provides a complementary methodological reference point. Panda et al. frame concept erasure as a variational inference problem, encouraging the posterior over model parameters to move away from regions associated with the target concept [17]. In a related line of work, concept erasure in text-to-image diffusion models has been implemented by fine-tuning attention weights to weaken associations between prompts and visual features [7]. Although these approaches differ from the VAE setting, they raise the same central evaluation issue: behavioural success does not necessarily imply internal forgetting. A model may stop producing the concept under standard prompts while still retaining recoverable representations under latent perturbations or alternative probes.

Altogether, it seems that concept unlearning is easier to demonstrate at the output level than to verify internally. That distinction motivates this project, focusing on how deeply a VAE can forget a visual concept by measuring changes in its latent manifold after different unlearning algorithms, and by comparing those changes against an oracle.

3.3 Representation Geometry and Evaluation Tools

3.3.1 The VAE Latent Manifold

A core theoretical assumption of this project is that a VAE trained on CelebA learns a structured latent manifold where semantically similar faces occupy nearby regions of latent space, and that visual attributes such as glasses correspond to identifiable submanifolds or directions. This assumption is grounded in well-established theory. Mathieu et al. showed that the latent space has a natural Riemannian metric induced by the decoder, which can be estimated from the data. Under this metric, semantically meaningful paths through latent space correspond to geodesics, and visual concepts correspond to compact regions of bounded intrinsic curvature [15].

The practical implication for unlearning is geometric: if a concept is encoded as a compact, well- separated submanifold in latent space, strong unlearning dissolves that submanifold — reducing its compactness, increasing its distance from the rest of the distribution, or reducing its effective dimensionality. Weak surface-level unlearning leaves the submanifold largely intact, with only the decoder output in that region modified. My latent space metrics 4.2 are chosen to detect these geometric changes.

3.3.2 Centered Kernel Alignment (CKA)

Centered Kernel Alignment (CKA), introduced by Kornblith et al. in "Similarity of Neural Network Representations Revisited", is the primary tool I will use to measure how deeply unlearning methods alter the VAE's internal representations. CKA addresses a fundamental problem with earlier representation similarity methods such as Canonical Correlation Analysis (CCA): CCA and related statistics are invariant to invertible linear transformations, and cannot measure meaningful similarities between representations of higher dimension than the number of data points [13].

CKA resolves this by operating on representational similarity matrices (RSMs) rather than on representations directly. For two sets of representations X and Y (e.g., the encoder's output at a given layer for glasses images, under the original vs. the unlearned model), CKA computes the normalised Hilbert—Schmidt Independence Criterion (HSIC) between their centered Gram matrices: $\mathrm{CKA}(X, Y) = \frac{\mathrm{HSIC}(K_t, L_t)}{\sqrt{\mathrm{HSIC}(K_t, K_t) \cdot \mathrm{HSIC}(L_t, L_t)}}$. The result lies in $[0, 1]$, with 1 indicating that the two representation sets are related by an orthogonal transformation and isotropic scaling (effectively identical structure), and 0 indicating no linear relationship between their similarity structures [13].

A key practical advantage is that CKA is invariant to orthogonal transformations and isotropic scalings but sensitive to the relational structure of representations. It detects when the model's internal topology of glasses images changes, regardless of which specific neurons encode the information. Kornblith et al. showed that CKA reliably identifies layer correspondences between networks trained from different random seeds, something CCA fails at.

In my setting, I compute layer-wise CKA between the original VAE encoder and each unlearned variant on a held-out test set of glasses images. A high CKA score on all encoder layers indicates that unlearning produced little internal change despite potentially different outputs — the hallmark of surface-level unlearning. A Low CKA score, particularly in deeper layers of the encoder, indicates genuine representational reorganisation. To ensure the metric reflects changes specific to the glasses submanifold, I compute CKA separately on glasses-only test batches rather than on the full face distribution.

3.3.3 Probing Classifiers

Linear probing tests whether a concept is decodable from frozen latent representations by a simple classifier. If a linear probe trained on the latent codes of a held-out test set can classify glasses vs. no-glasses with high accuracy, then the glasses concept is linearly decodable from those representations; the information remains, regardless of what the decoder outputs [8, 10]. Conversely, if probe accuracy drops to random-chance level after unlearning, the concept is not linearly recoverable from the latent space. The same procedure is applied to the dSprites setting, where heart vs. non-heart latents replace glasses vs. non-glasses test batches.

Golatkar et al. used probing functions to demonstrate that naive unlearning methods leave recoverable information in weights. Hsu et al. used probing to show that residual knowledge persists even when output-level metrics suggest successful unlearning. In my VAE setting, I can train a logistic regression probe on the encoder's latent means (the p vectors), using 80% of a held-out glasses/no-glasses test set for probe training and 20% for evaluation.

3.3.4 Adversarial Recovery

Adversarial recovery tests represent the most stringent evaluation of unlearning: if an adversary with access to the unlearned model and a glasses classifier can perturb latent codes to recover glasses-exhibiting outputs, then the unlearned model retains exploitable residual knowledge. This approach is inspired by Hsu et al.'s residual knowledge framework, showing that perturbed forget samples are recognised by inadequately unlearned models [10].

In my setting, I implement adversarial recovery as a latent-space attack. Starting from the latent code of a non-glasses face, I apply projected gradient descent guided by a pre- trained binary glasses classifier applied to the decoded output. I measure whether this optimisation can find a nearby latent point whose decoded image is classified as having glasses, and if so, how large a perturbation is required. A model for which small perturbations suffice retains a glasses manifold close to encoded non-glasses images; a model for which no perturbation within a large radius succeeds has successfully removed the glasses submanifold.

3.3.5 Dimensionality and Cluster Analysis

Cluster analysis and dimensionality reduction are classical tools for understanding representation space organisation that I adapt here for unlearning evaluation. Following prior work using clustering in latent space to detect forgotten categories [10, 16], I embed the encoder's latent means into 2D using t-SNE and UMAP, and compute the silhouette score of the resulting clusters. A high silhouette score indicates that forget-concept latent codes are tightly clustered and well-separated from retain-set codes; a low score after unlearning indicates genuine dispersal.

Following the geometric framework of Mathieu et al., I additionally compute k-NN graph connectivity measures, asking whether forget-concept latents form a connected neighbourhood in the full latent graph or become dispersed after unlearning [15]. Effective unlearning should fragment this structure and increase mixing with retain-set samples, indicating the concept no longer occupies a coherent region of latent space.

3.4 Synthesis and Open Challenges

Taken together, the literature converges on several key insights directly shaping my experimental design.

1. The distinction between surface-level and parametric unlearning is fundamental and empirically verifiable through internal representation analysis [8, 10]. Golatkar et al. demonstrated that weight-probing attacks recover forgotten information even when output behaviour matches the oracle; Hsu et al. extended this to the latent neighbourhood, showing residual knowledge persists near forget samples.

2. Feature unlearning in generative models has been demonstrated at the output level for attributes including glasses [16], but latent geometry has not been systematically examined. Sun et al.'s StyleGAN method and diffusion-model unlearning work by Panda et al. and Gandikota et al. independently confirm the same pattern: behavioural suppression does not imply internal forgetting [18, 17, 7].

3. CKA, probing, adversarial recovery, and cluster analysis provide complementary views of whether a concept has been genuinely erased, capturing representational structure, linear decodability, latent accessibility, and neighbourhood coherence respectively [13, 8, 10, 15].

4. Steering internal representations is more effective than output-level suppression alone [11], suggesting the most promising methods operate on latent geometry rather than decoder behaviour.

5. Residual knowledge is a persistent failure mode of approximate unlearning. In a VAE, where the latent space is a continuous manifold [15], a concept submanifold suppressed at the decoder can remain geometrically coherent in the encoder and accessible under latent-space attacks.

A major open challenge is the tension between unlearning completeness and model utility. Aggressive unlearning degrades retain-set performance [8], and unlearning images containing a target feature causes collateral damage to correlated attributes [16]. In my setting, glasses correlate with age and gender in CelebA, so including a β-VAE variant [9] allows me to test whether greater disentanglement limits this collateral damage. The dSprites setting, where generative factors are fully independent, provides a clean baseline for isolating this effect: any collateral damage observed there cannot be attributed to attribute correlations in the data.

A further challenge is the absence of a universal benchmark for generative unlearning. Unlike classifier unlearning, evaluating generative models requires jointly assessing perceptual quality, semantic fidelity, and concept absence [2, 3, 16]. My multi-metric framework—CKA, silhouette score, probe accuracy, adversarial recovery, k-NN connectivity, and reconstruction error—addresses this gap across two settings: dSprites, where ground-truth generative factors allow the framework to be validated in a controlled environment, and CelebA, where it is applied to the entangled, real-world case that Moon et al. identified as important but left unmeasured [16].

4. Project Plan

4.1 Research Questions

1. Do different unlearning methods produce different changes to the VAE's latent manifold geometry for the forgotten concept?
2. Do those geometric changes predict the degree of residual concept information; i.e., how well the concept can be recovered via probing or adversarial perturbation?
3. How closely does each method's effect on latent geometry approximate that of the oracle, and does intervention depth predict the size of that gap?

4.2 Hypotheses

- **H1 — Suppression vs. Erasure**
  Surface-level methods (output masking, decoder-only fine-tuning) will produce negligible change to encoder representations of forget-set inputs, yielding near-unity CKA with the original model and probe accuracy near the pre-unlearning baseline.
- **H2 — Geometry Predicts Recoverability**
  Methods that produce larger geometric disruption to the forget-set submanifold (lower CKA, collapsed cluster silhouette, reduced intrinsic dimension) will also produce lower probe accuracy and lower adversarial recovery rates, establishing a correlation between internal geometry and recoverability.
- **H3 — Oracle Approximation**
  The oracle will show maximum geometric deviation from the original for forget-set inputs, and minimum probe accuracy and adversarial recovery. Methods that most closely approximate the oracle's geometry will also most closely approximate it behaviourally.

4.3 Project Timeline and Milestones

- 20 Mar – 23 Apr 2026: **Setup.** Initial dataset preparation, baseline VAE implementation in PyTorch. *[Prerequisite for all RQs]*
- 24 Apr – 30 Apr 2026: **Baseline model and metrics.** Train VAE, extract latent representations, and implement probing classifiers. *[Establishes pre-unlearning baseline for RQ2, RQ3]*
- 1 May – 7 May 2026: **Analysis tools.** Implement PCA, t-SNE, or UMAP for latent space visualisation. *[Supports geometric measurement in RQ1, RQ2]*
- 8 May – 15 May 2026: **Core unlearning.** Implement oracle retraining and fine-tuning in preparation for the progress seminar. *[Addresses RQ3: establishes oracle reference geometry]*
- 16 May – 15 Jun 2026: **Core methods.** Complete negative gradient ascent and feature unlearning, with full evaluation. *[Addresses RQ1: compares geometric change across methods; RQ2: tests geometry–recoverability correlation]*
- 16 Jun – 15 Jul 2026: **Advanced methods.** Implement CKA and Fisher-based or representation-based approaches. *[Addresses RQ1, RQ3: deeper structural comparison against oracle]*
- 16 Jul – 21 Aug 2026: **Thesis plan.** Prepare and submit thesis plan with refined methodology and preliminary results. *[Consolidates findings across RQ1–RQ3]*
- 22 Aug – 30 Sep 2026: **Final experiments.** Perform hyperparameter tuning and finalise best-performing methods. *[Refines RQ2 correlation analysis; addresses RQ3 oracle approximation gap]*
- 1 Oct – 23 Oct 2026: **Poster and demonstration.** Prepare poster, demo, and visual summaries. *[Communicates conclusions of RQ1–RQ3]*
- 24 Oct – 9 Nov 2026: **Thesis writing.** Complete and submit final thesis report.

4.3.1 Key Milestones

- 13 May: Progress seminar
- 21 Aug: Thesis plan due
- 19 - 23 Oct: Poster and Demonstration
- 9 Nov: Thesis Report due

4.3.2 Progress to Date

As of submission (23 April 2026), I am at the end of the setup phase (20 Mar – 23 Apr). I have downloaded the CelebA and dSprites datasets locally and configured the core Python environment with PyTorch, torchvision, and scikit-learn. I have conducted an initial inspection of both datasets: I examined CelebA's attribute label file to verify the eyeglasses column and its class distribution, and reviewed the dSprites latent factor structure to confirm that the heart shape label is cleanly separable by generative factor. My next step is constructing the forget/retain splits and implementing the baseline VAE, scheduled for the baseline phase beginning 24 April.

4.3.3 Resource Allocation per Phase

Table 4.1 summarises the primary compute, data, and software dependencies for each project phase.

Phase	**Est. Compute**	**Data / Storage**	**Software / Dependencies**
Setup (20 Mar – 23 Apr)	<5 GPU hrs (local)	CelebA ∼1.4 GB; dSprites ∼80 MB	PyTorch 2.x, torchvision, NumPy, Matplotlib
Baseline model & metrics (24 Apr – 30 Apr)	∼10 GPU hrs (UQ HPC)	Preprocessed splits cached on HPC scratch	scikit-learn, tqdm, wandb (logging)
Analysis tools (1 May – 7 May)	<2 GPU hrs (local)	Saved latent vectors (∼500 MB per model)	umap-learn, sklearn.manifold, Matplotlib
Core unlearning (8 May – 15 May)	∼20 GPU hrs (UQ HPC)	Retain/forget split indices; oracle checkpoint	PyTorch training loop, torchvision
Core methods (16 May – 15 Jun)	∼40 GPU hrs (UQ HPC)	All checkpoints ∼4 GB total	scikit-learn (probing), Moon et al. attribute classifier
Advanced methods (16 Jun – 15 Jul)	∼20 GPU hrs (UQ HPC)	Cached representations for CKA	torch-cka or custom NumPy CKA implementation
Final experiments (22 Aug – 30 Sep)	∼30 GPU hrs (UQ HPC)	Full result logs; hyperparameter sweep outputs	Optuna or manual grid search, wandb
Thesis writing (24 Oct – 9 Nov)	Nil	Final figures exported as PDF/PNG	LaTeX, Overleaf

Table 4.1: Resource allocation per project phase. UQ HPC refers to the Bunya cluster; Google Colab Pro is available as a backup if HPC queues cause delays.

4.4 Methodology and Experimental Design

Unlearning methods are organised along two axes: **feasibility tier** (primary, secondary, stretch goal) and **intervention depth**. Intervention depth is the degree of internal forgetting, with naive methods producing negligible geometric change and weight-level methods most closely approximating the oracle. Tables 4.2 and 4.3 summarise all methods.

- **Naive methods** — retain-set fine-tuning and negative gradient ascent make no attempt to target internal representations. Both are included because prior work consistently shows they leave residual information recoverable under probing [8], establishing the lower bound against which stronger methods are compared.
- **Representation-level methods** — Moon et al.'s feature unlearning and Huu-Tien et al.'s representation misdirection explicitly manipulate where the concept lives in latent space, making them the most directly relevant to this project's evaluation framework [16, 11].
- **Weight-level methods** — Fisher scrubbing and influence-guided unlearning use second-order approximations to push the model's weight distribution toward the oracle's. They offer principled guarantees absent from the other tiers, but their adaptation to the VAE ELBO objective is itself an open question.

All experiments run on CelebA (eyeglasses) as the primary setting and dSprites (heart shapes) as a controlled validation, across two model variants: a standard VAE ($\beta = 1$) and a β-VAE ($\beta = 4$). The dSprites setting isolates any benefit of disentanglement from the attribute correlations present in CelebA; reconstruction error is interpreted relative to each variant's own baseline, since β-VAE outputs are typically blurrier by design.

Priority	Method	Description and Implementation	Justification
Primary	Dimensionality & Cluster Analysis	Compute latent vectors z and visualise structure using PCA, t-SNE, or UMAP via `sklearn` or `umap-learn`, with plots generated in `matplotlib`.	These methods provide complementary views of latent geometry: PCA captures global linear structure; t-SNE and UMAP reveal nonlinear clustering. This detects whether unlearning disperses the forget-concept rather than outputs.
Primary	Probing Classifiers	Extract latent vectors from a PyTorch VAE and train a linear probe (logistic regression or small MLP) on (z, label) using `scikit-learn` or PyTorch.	Probing tests whether the feature remains decodable from latent representations, addressing the failure mode where output behaviour appears correct but internal representations still encode the concept. A linear probe is interpretable and consistent with prior work.
Primary	k-NN Graph Connectivity	Construct a k-NN graph over latent vectors z (e.g. using `scikit-learn`) and measure connectivity, cluster purity, and mixing between classes before and after unlearning.	Following the geometric perspective of [15], connectivity in latent neighbourhood graphs provides a proxy for whether a concept forms a coherent manifold. Effective unlearning should disperse the forget-concept cluster, reducing connectivity and increasing mixing with retain-set samples.
Secondary	CKA	Compute CKA between latent representations before and after unlearning using NumPy, PyTorch, or a `torch-cka` implementation.	CKA measures structural similarity between representations independent of neuron alignment, another method for detecting internal changes to the latent manifold. It complements probing by capturing geometric change rather than decodability.
Stretch	Adversarial Recovery	Attempt to recover the removed feature by training a nonlinear classifier on z or optimising latent vectors via gradient ascent in PyTorch.	This provides the strongest test of unlearning by an adversary attempting through building perturbations to attempt a recovery of the feature. It goes beyond passive evaluation (e.g. probing) and is particularly relevant for continuous latent spaces where nearby regions may still encode the concept.

Table 4.2: Evaluation methods for measuring residual information in latent representations.

Priority	Method	Description and Implementation	Justification
Primary	Oracle Retraining	Filter the dataset to remove the forget-concept (eyeglasses in CelebA; hearts in dSprites) and train a VAE from scratch on the retain set using PyTorch.	Provides the gold-standard baseline for exact unlearning. It represents the model obtained without the forget set. All approximate methods are evaluated relative to this reference.
Primary	Fine-tuning on Retain Set	Continue VAE training on the retain dataset only using a standard PyTorch training loop.	A simple, commonly used baseline. It evaluates of the gap between naive behavioural forgetting and true parametric unlearning, as prior work shows it often leaves residual information.
Primary	Negative Gradient Ascent	Maximise loss on the forget set via $L = L_{\text{retain}} - \alpha L_{\text{forget}}$ implemented in PyTorch.	A direct parametric approach to unlearning with low implementation complexity. Although known to be imperfect, it still tests whether explicitly pushing away forget samples improves over naive fine-tuning.
Primary	Feature Unlearning (Moon et al.)	Identify the latent direction corresponding to the feature via a linear probe and remove it using $z - (z \cdot w)w$, or penalise it during training.	Specifically designed for feature-level unlearning in generative models, making it highly tailored to this project, bridging the gap between output-level suppression and latent-space manipulation.
Secondary	Representation Misdirection (Huu-Tien et al.)	Add a loss encouraging z_{forget} to move toward random latent vectors (e.g. MSE) during training.	Inspired by recent work suggesting that modifying internal representations may be more effective than loss maximisation alone. It directly targets latent geometry, making it a creative extension.
Stretch	Influence-guided Unlearning (Fan et al.)	Approximate parameter updates for removing forget samples using influence functions, implemented via scaled gradient ascent on the forget set without full retraining.	Provides a computationally efficient approximation to retraining by estimating each sample's effect on model parameters. It offers a more principled alternative to naive gradient ascent, though adapting influence estimation to the VAE ELBO objective introduces additional complexity.
Stretch	Fisher Information Scrubbing (Golatkar et al.)	Approximate Fisher Information via squared gradients and inject parameter noise scaled by its inverse in a custom PyTorch implementation.	A theoretically grounded approach targeting information encoded in model weights, addressing limitations of output-level methods. Test whether principled approaches better approximate the oracle.

Table 4.3: Unlearning methods evaluated in this project.

4.5 Risk Assessment

4.5.1 OHS Risk Assessment

This project is conducted entirely in a low-risk computing environment where laboratory, field, or workshop work is not required. All experiments are performed computationally using a personal workstation and/or UQ cloud computing resources. The primary physical risks are musculoskeletal strain from prolonged sitting and eye strain from screen use. These are mitigated by adhering to ergonomic practices, taking regular breaks, and using appropriate display settings.

4.5.2 Ethics Assessment

This project does not involve human participants, animals, or the collection of personal data. The datasets used, CelebA and dSprites, are publicly available benchmarks released for academic research. CelebA consists of celebrity face images that are publicly available and widely used in the machine learning literature; no individual is identifiable in the context of this project, as images are used solely as training data for generative model experiments. No UQ ethical clearance is required.

4.5.3 Project Risks

The following risks to project schedule and deliverability have been identified:

Risk	Likelihood	Impact	Mitigation
Insufficient compute for CelebA VAE training (GPU memory, training time)	Medium	Major	Use UQ HPC cluster or Google Colab Pro as backup. Reduce image resolution or latent dimensionality if needed. Schedule training runs early to allow for reruns.
Fisher scrubbing or influence-guided unlearning proves infeasible to adapt to the VAE ELBO objective within the timeline	High	Minor	Both methods are designated stretch goals. If adaptation proves intractable, the project remains complete with primary and secondary methods. Effort is redirected to deeper evaluation of implemented methods.
Feature Unlearning (Moon et al.) fails to converge or produces degenerate reconstructions	Medium	Moderate	Two variants are planned (projection and penalised fine-tuning); if one fails, the other is attempted. Hyperparameter tuning is scheduled in the final experiments phase (22 Aug – 30 Sep).
CelebA glasses class imbalance degrades probe or clustering metrics	Medium	Moderate	Apply stratified sampling when constructing forget/retain splits. Report class-balanced accuracy for all probing evaluations.
dSprites experiments reveal no meaningful difference between VAE and β-VAE unlearning, weakening the comparative narrative	Low	Minor	A null result is itself a valid and publishable finding. Frame as evidence that disentanglement does not facilitate unlearning in the idealised case, which strengthens rather than undermines the thesis contribution.
Timeline slippage due to underestimating implementation complexity of CKA (PyTorch hooks)	Medium	Minor	CKA is designated secondary priority. If implementation is delayed, primary metrics (probing, clustering, k-NN) are sufficient for the progress seminar. CKA is completed in the advanced methods phase (16 Jun – 15 Jul).

Table 4.4: Project risk assessment.

5. AI Use Statement

I acknowledge the use of generative AI tools in completing this assessment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content, and ensuring it adheres to academic integrity standards.

Have I used AI to

- Explore your topic? Yes/No. Explain how (or explain why not):
  Yes. I used ChatGPT (GPT-4o) to assist in refining my understanding of machine unlearning, particularly in the context of latent representations and evaluation methods. It suggested relevant subtopics and terminology, which I then independently verified through academic sources, Google Scholar and conference proceedings.

- Define your research question? Explain how (or explain why not).
  No. The research question was formulated independently. AI-generated suggestions were considered but were generally too broad or not sufficiently aligned with the intended scope, and were therefore not used.

- Assist with your literature review? Explain how (or explain why not).
  Yes. ChatGPT was used to help filter and prioritise relevant literature. For example, I prompted it to identify papers related to machine unlearning in VAEs and then refine the list based on relevance and publication venue (e.g., ICML, ICLR, NeurIPS). I independently verified all selected papers and used my own judgement in determining their relevance.

- Assist with formatting the answers presented here? (e.g., grammar, spelling, language) Explain how (or explain why not).
  Yes. AI was used selectively to improve clarity and structure in dense sections. For example, I provided rough notes or bullet points and asked ChatGPT to consolidate them into coherent paragraphs. These outputs were then rewritten and edited in my own words, with citations manually inserted in LaTeX.

- Other aspects of your work? Explain how you used it.
  Yes. ChatGPT was used extensively for technical and formatting support. This included debugging LaTeXissues (e.g., table formatting, spacing, and layout adjustments), structuring sections of the document, and refining presentation elements such as tables and timelines. It also found inconsistencies in this documents template, and it taught me latex skills. All suggestions were reviewed and selectively incorporated.

AI Model and date	Table 1. AI tool usage								
	Language Translation	Grammar / style / spelling	Topic exploration	Research question	Literature review	Content Creation visual (formatting)	Content Creation (code, text)	Feedback	Other
ChatGPT 4o 09/03 – 14/04		✓	✓		✓	✓	✓	✓	LaTeX debugging, document structuring

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